

Municipal Councils E-Voting System

Software Requirements Engineering Project

CESW221

Group Members:

Taif Mabruk 2410500

Raghad Turki 2410824

Rmaas Badr 2411185

Reem Aljuhani 2412376

Table of Contents

Task 1: Background and Goals of the Project

Task 2: Business Rules

Task 3: Stakeholder Map

Task 4: Context Diagram

Task 5: Product Use Case Diagram

Task 6: Use Case Scenarios

Task 7: Requirements Traceability Matrix

Task 8: Functional Requirements Using Volere Snow Card

Task 9: Non-Functional Requirements Using Volere Snow Card

Task 10: Prototype Design

Task 1: Background and Goals of the Project

Project Background

The Municipal Councils in Saudi Arabia currently use traditional voting methods, and the existing system does not support e-voting. Because of the low participation rate in municipal elections, a new e-voting system has been proposed as an additional voting channel.

The purpose of this system is not to replace classical or postal voting, but to make voting easier and more accessible through the Internet. The system should allow eligible citizens to vote online after verifying their identity using GOV.SA / NAFATH. It should also ensure confidentiality, security, and that each voter votes only once.

This project aims to support public participation, especially among young adults who frequently use the Internet, while still protecting the rights of citizens who do not have Internet access.

Project Goals

1. To develop an easy and practical e-voting system for municipal council elections.
2. To increase citizens' participation by providing an additional online voting channel without replacing traditional voting methods.
3. To verify voters' identities using GOV.SA / NAFATH and ensure that only eligible citizens can vote.
4. To ensure that each voter votes only once and that all submitted votes are counted correctly.
5. To protect the confidentiality and privacy of voters' choices while providing a secure and reliable voting system.

Task 2: Business Rules

The following business rules define the main constraints of the e-voting system, including voter eligibility, one vote per voter, confidentiality, vote counting, identity verification through NAFATH, and compliance with municipal council laws.

BR-ID	Business Rule	Source
BR-1	Only eligible voters may take part in the election.	Legal provisions
BR-2	Each voter must have only one vote.	Legal provisions
BR-3	Each voter must vote only once.	Legal provisions
BR-4	The system shall use GOV.SA / NAFATH to identify citizens.	National Council
BR-5	The voter's choice must remain confidential and cannot be revealed without permission.	Legal provisions
BR-6	All submitted votes must be counted correctly.	Legal provisions
BR-7	Citizens who do not have Internet access must not be penalized.	National Council
BR-8	The system must comply with municipal councils laws and regulations issued by royal decree.	National Council

Task 3: Stakeholder Map

The stakeholder map identifies the main stakeholders involved in the Municipal Councils E-Voting System and classifies them based on their level of involvement: directly involved, directly supporting, and indirectly influencing.

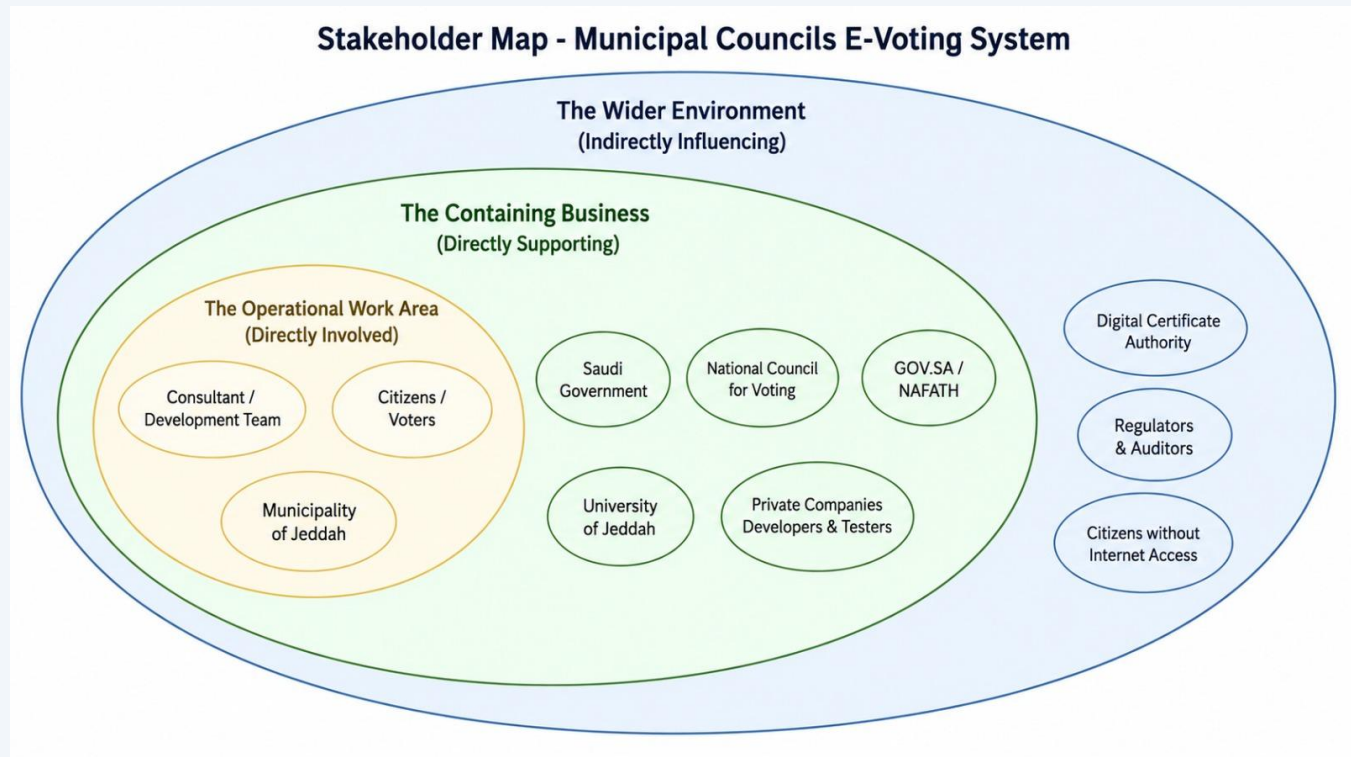


Figure 3.1: Stakeholder Map

The operational work area includes the direct users and development stakeholders, such as Citizens/Voters, Municipality of Jeddah, and the Consultant/Development Team. The containing business includes supporting and governing organizations such as the Saudi Government, National Council for Voting, GOV.SA/NAFATH, University of Jeddah, and private companies. The wider environment includes external parties that influence the system indirectly, such as regulators, auditors, digital certificate authorities, and citizens without Internet access.

Task 4: Context Diagram

The context diagram shows the E-Voting System as one central system and illustrates its interactions with the main external entities.

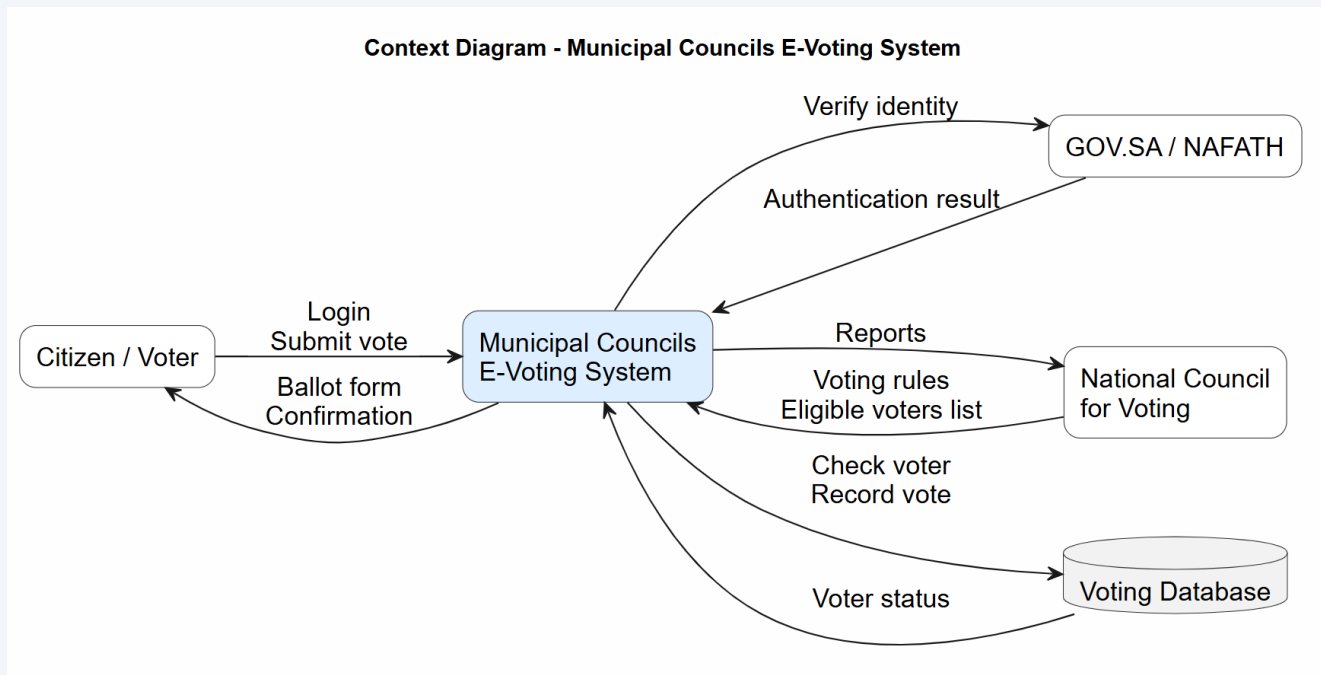


Figure 4.1: Context Diagram

In this diagram, the voter logs in and submits the vote through the system. The system verifies the voter's identity using GOV.SA / NAFATH, receives voting rules and eligible voters list from the National Council, and uses the Voting Database to check voter status and record the vote.

Task 5: Product Use Case Diagram

The product use case diagram shows the main functions of the Municipal Councils E-Voting System. The system is divided into three main areas: Authentication, Voting Process, and Administration.

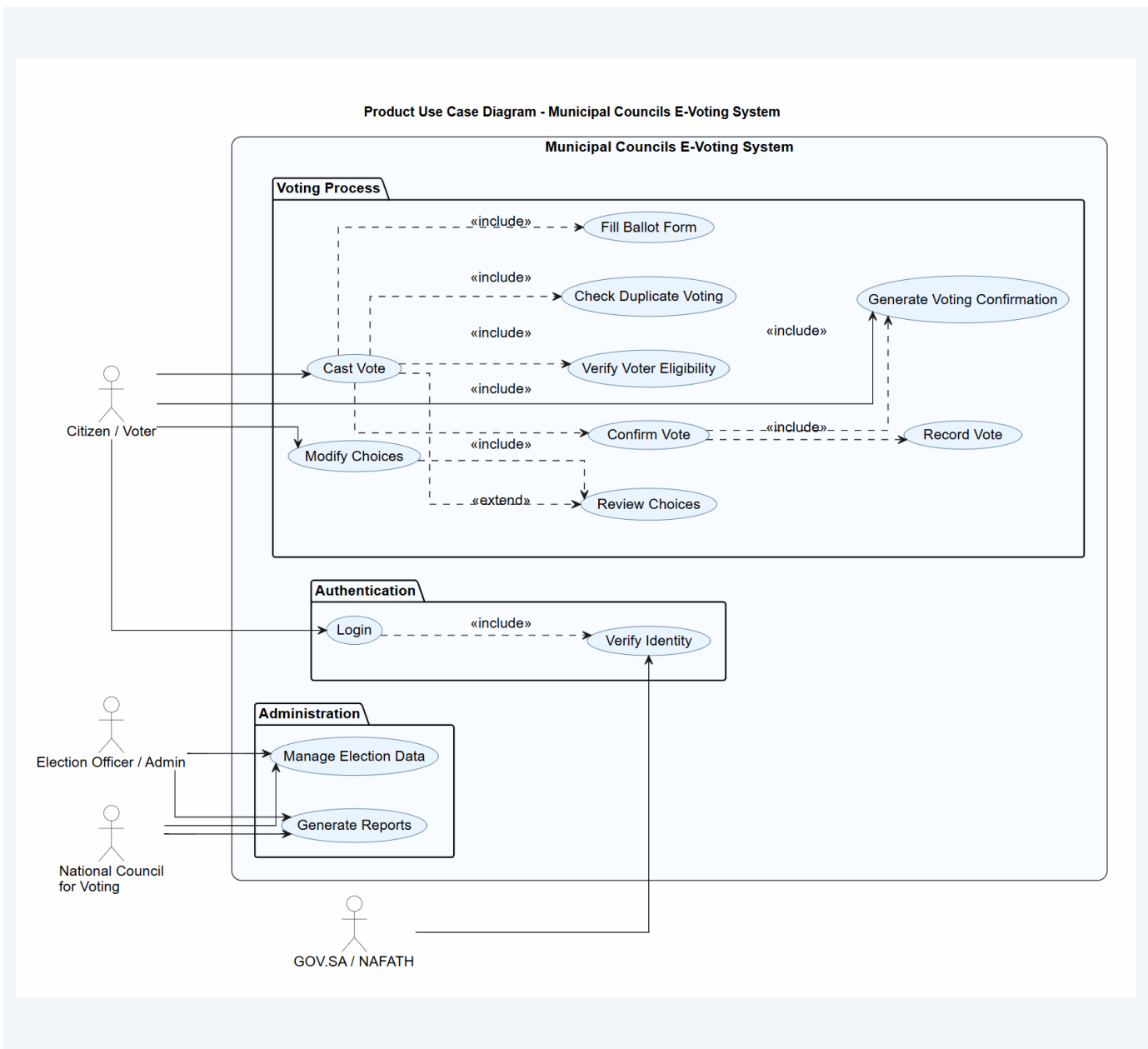


Figure 5.1: Product Use Case Diagram

The voter can log in, cast a vote, fill the ballot form, review or modify choices, confirm the vote, and receive confirmation. The system also verifies identity through GOV.SA / NAFATH, checks voter eligibility, prevents duplicate voting, records the vote, and allows election officers and the National Council to manage election data and generate reports.

Task 6: Use Case Scenarios

The following use case scenarios describe two important use cases in the Municipal Councils E-Voting System: Login and Cast Vote. These scenarios show the actors, goals, preconditions, main flows, alternative flows, exception flows, and postconditions.

Use Case Scenario 1: Login

Field	Description
Use Case Name	Login
Primary Actor	Citizen / Voter
Supporting Actor	GOV.SA / NAFATH
Goal	To allow the voter to access the e-voting system after verifying their identity.
Preconditions	The voter has access to the e-voting system and has valid GOV.SA / NAFATH credentials.
Trigger	The voter opens the e-voting system and selects the login option.
Main Flow	1. The voter opens the Municipal Councils E-Voting System. 2. The voter selects the login option. 3. The system redirects the voter to GOV.SA / NAFATH. 4. The voter enters the required identity information. 5. GOV.SA / NAFATH verifies the voter's identity. 6. GOV.SA / NAFATH sends the authentication result to the e-voting system. 7. The system allows the voter to continue to the voting process.
Alternative Flow	If the voter enters incorrect information, the system allows the voter to try again.
Exception Flow	If identity verification fails, the system denies access and displays an error message.
Postconditions	The voter is authenticated and can proceed to the voting process.

Use Case Scenario 2: Cast Vote

Field	Description
Use Case Name	Cast Vote
Primary Actor	Citizen / Voter
Supporting Actors	Voting Database, National Council for Voting
Goal	To allow an eligible voter to submit one electronic vote successfully.
Preconditions	The voter is logged in and authenticated through GOV.SA / NAFATH.
Trigger	The voter selects the option to start voting.
Main Flow	1. The voter selects "Cast Vote." 2. The system verifies the voter's eligibility. 3. The system checks that the voter has not voted before. 4. The system displays the online ballot form. 5. The voter fills in the ballot form by selecting the preferred choices. 6. The system displays a summary of the voter's choices. 7. The voter reviews the choices. 8. The voter confirms the vote. 9. The system records the vote in the voting database. 10. The system marks the voter as having voted. 11. The system generates a voting confirmation with the date and time. 12. The system displays the confirmation to the voter.
Alternative Flow	If the voter wants to change the choices, the voter selects "Modify Choices." The system returns the voter to the ballot form. The voter updates the choices, reviews the updated summary, and then confirms the vote.
Exception Flow	If the voter is not eligible, the system stops the voting process and displays an error message. If the voter has already voted, the system rejects the voting attempt and displays a message that the voter has already submitted a vote.
Postconditions	The vote is recorded successfully, and the voter is marked as having voted.

Task 7: Requirements Traceability Matrix

The following requirements traceability matrix links the functional requirements to their sources, including use cases and business rules. The priorities are assigned using the MoSCoW prioritization technique.

FR ID	Functional Requirement Description	FR Source	FR Priority	Related Non-Functional Requirement
FR-1	The system shall allow citizens to log in using GOV.SA / NAFATH.	UC-1 Login, BR-4	Must have	Security, Usability
FR-2	The system shall verify the voter's identity before allowing access to the voting process.	UC-1 Login, BR-4	Must have	Security
FR-3	The system shall verify that only eligible voters can participate in the election.	UC-2 Cast Vote, BR-1	Must have	Security, Reliability
FR-4	The system shall ensure that each voter has only one vote.	UC-2 Cast Vote, BR-2	Must have	Integrity, Security
FR-5	The system shall prevent a voter from voting more than once.	UC-2 Cast Vote, BR-3	Must have	Integrity, Security
FR-6	The system shall display an online ballot form to the voter.	UC-2 Cast Vote	Must have	Usability
FR-7	The system shall allow the voter to review and modify choices before final confirmation.	UC-2 Cast Vote	Should have	Usability
FR-8	The system shall record the confirmed vote in the voting database.	UC-2 Cast Vote, BR-6	Must have	Reliability, Integrity
FR-9	The system shall generate a voting confirmation with date and time after successful submission.	UC-2 Cast Vote	Must have	Usability, Auditability
FR-10	The system shall keep the voter's choice confidential and prevent unauthorized disclosure.	BR-5	Must have	Privacy, Confidentiality, Security
FR-11	The system shall not penalize citizens who do not have Internet access by keeping alternative voting channels available.	BR-7	Must have	Accessibility, Fairness
FR-12	The system shall comply with municipal council laws and regulations issued by royal decree.	BR-8	Must have	Compliance

Task 8: Functional Requirements Using Volere Snow Card

The following functional requirements are written using the Volere Snow Card template. These requirements were selected from the requirements traceability matrix because they represent core functions of the e-voting system.

Functional Requirement 1

Field	Description
Requirement #	FR-1
Requirement Type	Functional Requirement
Event / Use Case #	UC-1 Login
Description	The system shall allow citizens to log in using GOV.SA / NAFATH.
Rationale	To verify the voter's identity before accessing the e-voting system.
Originator	National Council for Voting
Fit Criterion	A citizen with valid GOV.SA / NAFATH credentials shall be able to log in successfully.
Customer Satisfaction	5
Customer Dissatisfaction	5
Priority	Must have
Supporting Materials	Case Study, Business Rule BR-4
Conflicts	Citizens without access to GOV.SA / NAFATH may need alternative voting channels.
History	Created during the requirements analysis phase.

Functional Requirement 2

Field	Description
Requirement #	FR-5
Requirement Type	Functional Requirement
Event / Use Case #	UC-2 Cast Vote
Description	The system shall prevent a voter from voting more than once.
Rationale	To ensure fairness and integrity in the election process.
Originator	Legal Provisions / National Council for Voting
Fit Criterion	If a voter ID is already marked as voted, the system shall reject any new voting attempt.
Customer Satisfaction	5
Customer Dissatisfaction	5
Priority	Must have
Supporting Materials	Case Study, Business Rule BR-3
Conflicts	The system must check voting status across all voting channels, including online, postal, and polling station voting.
History	Created during the requirements analysis phase.

Functional Requirement 3

Field	Description
Requirement #	FR-9
Requirement Type	Functional Requirement
Event / Use Case #	UC-2 Cast Vote
Description	The system shall generate a voting confirmation with date and time after successful vote submission.
Rationale	To inform the voter that the vote has been successfully recorded.
Originator	Citizen / Voter
Fit Criterion	After a successful vote submission, the system shall display a confirmation message including the date and time of recording.
Customer Satisfaction	4
Customer Dissatisfaction	4
Priority	Must have
Supporting Materials	Case Study, Use Case Scenario: Cast Vote
Conflicts	The confirmation must not reveal the voter's selected choices to protect confidentiality.
History	Created during the requirements analysis phase.

Task 9: Non-Functional Requirements Using Volere Snow Card

This section presents three non-functional requirements: security, privacy, and usability.

Non-Functional Requirement 1

Field	Description
Requirement #	NFR-1
Requirement Type	Non-Functional Requirement – Security
Event / Use Case #	UC-1 Login, UC-2 Cast Vote
Description	The system shall protect the communication between the voter's device and the e-voting server using secure encryption.
Rationale	To protect voter information and voting data from unauthorized access during transmission.
Originator	National Council for Voting
Fit Criterion	All communication between the voter's device and the e-voting server shall be encrypted using SSL or a stronger security protocol.
Customer Satisfaction	5
Customer Dissatisfaction	5
Priority	Must have
Supporting Materials	Case Study security requirements
Conflicts	Strong security measures may make the system slower or harder to use for some citizens.
History	Created during the requirements analysis phase.

Non-Functional Requirement 2

Field	Description
Requirement #	NFR-2
Requirement Type	Non-Functional Requirement – Privacy / Confidentiality
Event / Use Case #	UC-2 Cast Vote
Description	The system shall ensure that the voter's identity cannot be linked to the voter's selected choices without permission.
Rationale	To protect the confidentiality of the vote and increase public trust in the e-voting system.
Originator	Legal Provisions / Municipality of Jeddah
Fit Criterion	No unauthorized user shall be able to retrieve a link between the voter's identity and the voter's ballot choices.
Customer Satisfaction	5
Customer Dissatisfaction	5
Priority	Must have
Supporting Materials	Case Study, Business Rule BR-5
Conflicts	The system still needs to verify voter eligibility while keeping the vote confidential.
History	Created during the requirements analysis phase.

Non-Functional Requirement 3

Field	Description
Requirement #	NFR-3
Requirement Type	Non-Functional Requirement – Usability
Event / Use Case #	UC-1 Login, UC-2 Cast Vote
Description	The system shall be easy and practical for citizens to use during the voting process.
Rationale	To encourage citizens to participate in municipal elections and reduce the need for training.
Originator	National Council for Voting / Municipality of Jeddah
Fit Criterion	A first-time voter shall be able to complete the login and voting process with clear instructions and minimal assistance.
Customer Satisfaction	4
Customer Dissatisfaction	4
Priority	Should have
Supporting Materials	Case Study, Municipality of Jeddah letter
Conflicts	Extra security and authentication steps may reduce ease of use.
History	Created during the requirements analysis phase.

Task 10: Prototype Design

The prototype design includes both low-fidelity and high-fidelity screens for the Municipal Councils E-Voting System. The low-fidelity prototype shows the initial hand-drawn layout, while the high-fidelity prototype presents the final mobile interface design created in Figma.

Low-Fidelity Prototype

The low-fidelity prototype represents the initial design of the main system screens. It focuses on the basic structure, layout, and user flow without final colors, icons, or detailed visual elements.



Figure 10.1: Low-Fidelity Login Screen

This screen shows the basic login layout, where the voter starts the process by accessing the system and verifying their identity.

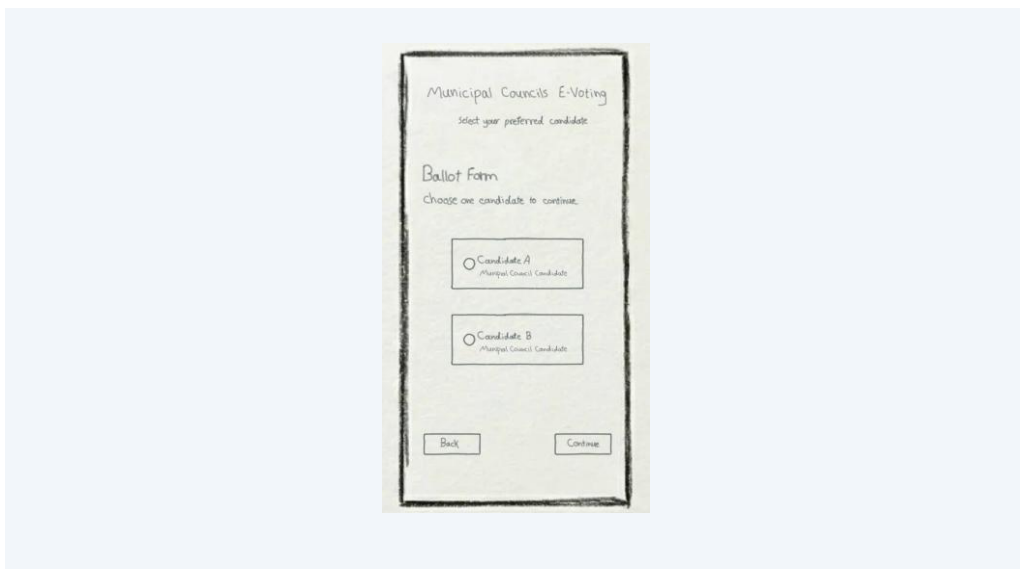


Figure 10.2: Low-Fidelity Ballot Form Screen

This screen shows the initial ballot form structure, where the voter can view the available candidates and select one candidate.



Figure 10.3: Low-Fidelity Confirmation Screen

This screen shows the basic confirmation page that appears after the vote is submitted successfully.

High-Fidelity Prototype

The high-fidelity prototype was created in Figma. It shows the final mobile interface design with colors, icons, buttons, spacing, and a more organized visual layout.

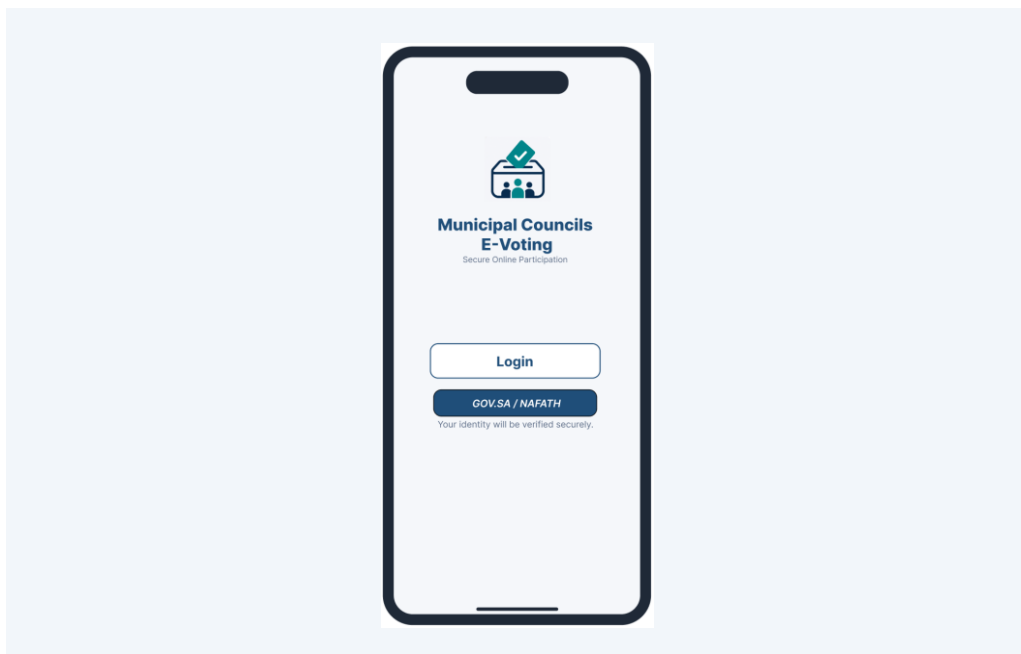


Figure 10.4: High-Fidelity Login Screen

This screen allows the voter to access the system and verify their identity through GOV.SA / NAFATH.

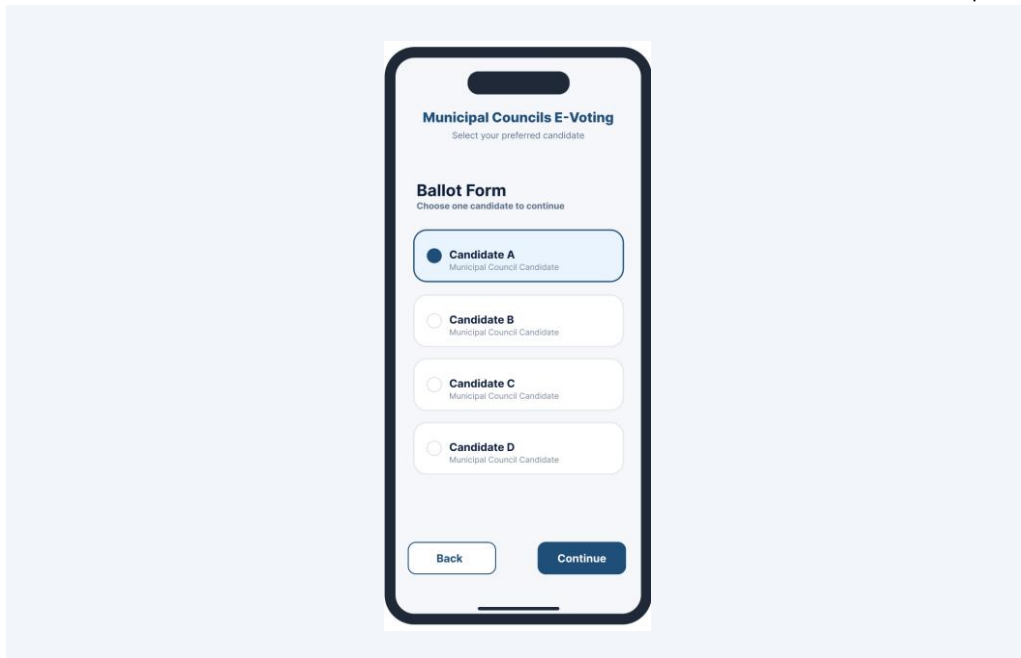


Figure 10.5: High-Fidelity Ballot Form Screen

This screen allows the voter to view the available candidates, select one candidate, and continue to the next step.

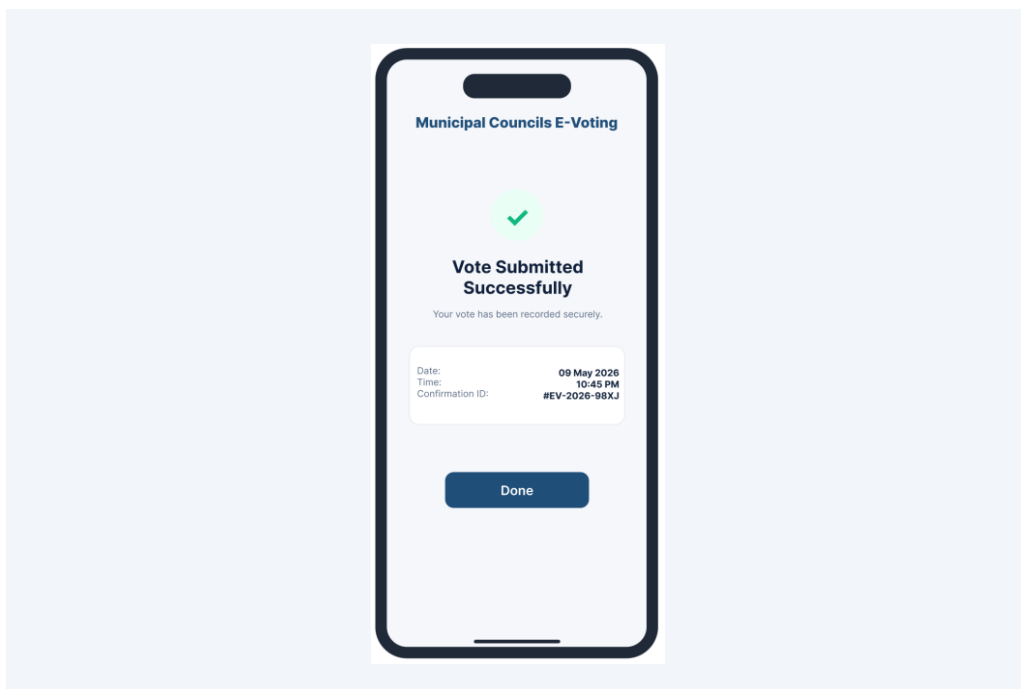


Figure 10.6: High-Fidelity Confirmation Screen

This screen confirms that the vote has been submitted successfully and displays the date, time, and confirmation ID.

References

Project Case Study: Municipal Councils E-Voting System in Saudi Arabia.

Project Tasks: CESW221 Software Requirements Engineering Project.